

EXP 20:

Perform Sharpening of Image using Laplacian mask with negative center coefficient.

0	1	0
1	-4	1
0	1	0

AIM:

Sharpening of Image using Laplacian mask with negative center coefficient.

Program:

```
import cv2

import numpy as np

img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

else:

    kernel = np.array([[0, 1, 0],
                       [1, -4, 1],
                       [0, 1, 0]])

    sharpened = cv2.filter2D(img, -1, kernel)

    cv2.imshow('Original Image', img)

    cv2.imshow('Sharpened Image', sharpened)

    cv2.imwrite('Sharpened_Image.jpg', sharpened)

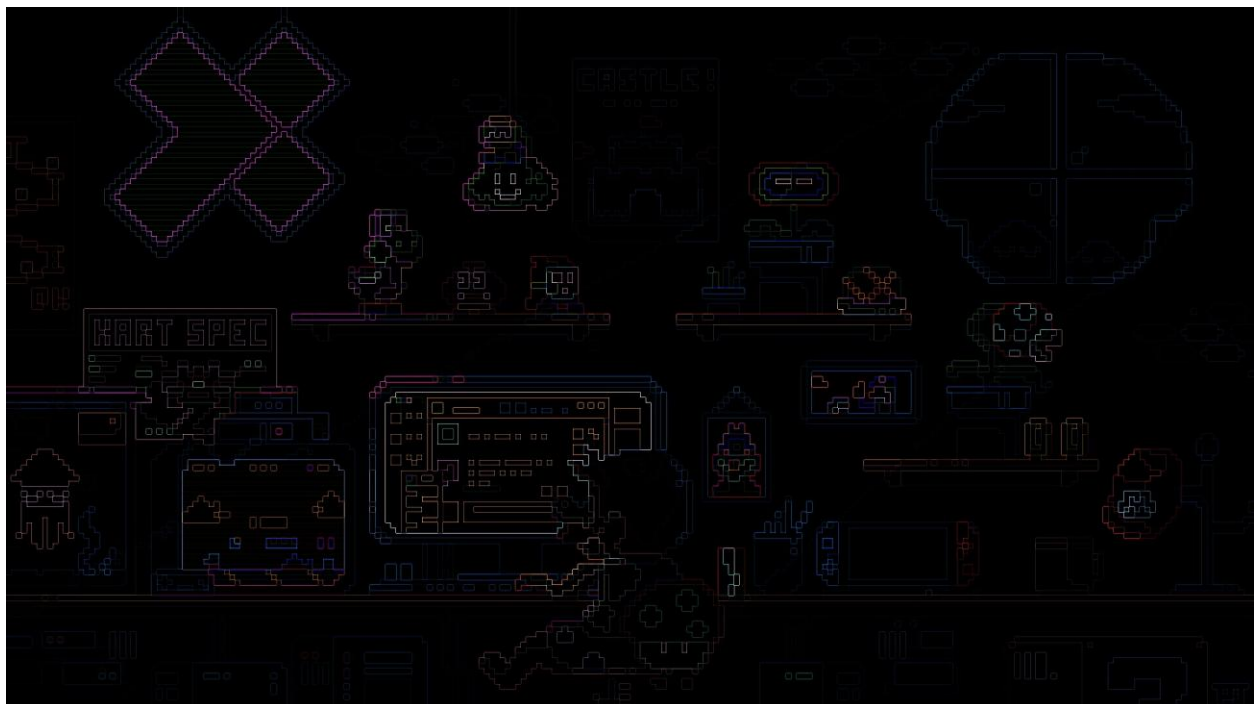
    cv2.waitKey(0)

    cv2.destroyAllWindows()
```

INPUT:



Output:



RESULT:- Thus the program is executed successfully.